Chapter 16: Capstone Project 1 — Small Molecule Oncology Target Discovery

Chapter 15 closed by handing off to this book's own Part V capstone sequence: two complete, real, end-to-end projects that chain many of the individual methods Chapters 1-15 each built and validated in isolation into a single working discovery pipeline. This chapter is the first: a real small-molecule discovery campaign against a real oncology target, from live bioactivity data all the way to a real, automatically generated technical report. Nothing in this chapter is methodologically new — every stage below is a real method this book has already introduced, implemented, and validated in an earlier chapter. What is new is chaining all of them together, for the first time, into one working pipeline, and the specific real engineering decisions that chaining requires: a QSAR model that must serve two different roles at once (an evaluation metric *and* a differentiable reward signal for a generative model), a generative model whose reward oracle is upgraded from Chapter 7′s own binary classifier to this chapter's continuous regressor, and a report generator that must assemble every real number and structure the other four stages produce into one coherent, real technical deliverable.

16.1 Objective

This capstone's real target is the same one this book has followed since Chapter 1: the epidermal growth factor receptor (EGFR) tyrosine kinase domain, ChEMBL target CHEMBL203, PDB structure 1M17 in complex with the real, clinically approved lung-cancer drug erlotinib (Stamos, Sliwkowski, & Eigenbrot, 2002). EGFR is a real, ATP-competitive kinase target in the specific, precise sense the outline's "competitive inhibitors" framing calls for: erlotinib and its real relatives bind directly in the ATP-binding cleft of the kinase domain, competing with ATP itself for occupancy — a mechanistically different, and considerably better-precedented, real inhibition mode than, for instance, KRAS's own real covalent/allosteric switch-II-pocket mechanism (a different, equally real, but not classically "competitive" oncology target this book does not use for that reason). Reusing EGFR here, rather than introducing a new target for its own sake, is a deliberate choice: it lets this capstone build directly on top of Chapter 11′s already-validated real receptor preparation and docking protocol and Chapter 7′s already-validated real generative pipeline for this exact target, rather than re-deriving either from scratch — the same "reuse validated real infrastructure rather than re-verify it" discipline Chapter 13 already applied when it reused Chapters 11-12′s docking and MD methods unchanged for a different real target.

The real objective, concretely: design novel, drug-like, EGFR-active small molecules — sequences never seen during training, distinct from every compound in the real ChEMBL training data — and carry a real, evidence-based shortlist of them all the way through computational verification (predicted activity, drug-likeness, docking pose, and short-timescale conformational stability), exactly the real decision a medicinal chemistry team would need before committing wet-lab synthesis resources to any one candidate.

16.2 Pipeline Execution

The project code lives in this chapter's folder (`ch16_capstone_oncology/oncology_capstone.py`). Five real stages run in sequence, each stage's real output feeding the next.

16.2.1 Stage 1 — Real data: ChEMBL bioactivity + PDB structure validation

`extract_bioactivities` and `clean_bioactivity_records` are Chapter 7's own real extraction/curation methodology, reused essentially unchanged (live, paginated ChEMBL REST API retrieval; structure standardization via Chapter 4's largest-fragment/uncharge/tautomer-canonicalization pipeline; deduplication by the real median IC50 across repeated measurements per compound). This chapter adds one real, necessary extension Chapter 7 did not need: a continuous **pIC50** regression target, $\mathrm{pIC50} = 9 - \log_{10}(\mathrm{IC50}_{\mathrm{nM}})$, computed alongside the existing binary active/inactive label, since Stage 2's QSAR model is a regressor, not (as Chapter 7's reward oracle was) a classifier. Before any docking is trusted downstream, the bundled real PDB 1M17 structure is validated directly — confirmed to contain both a real protein chain (2,511 real protein atoms) and the real, co-crystallized erlotinib (PDB ligand code AQ4, 29 real ligand atoms) HETATM block — the same basic, disclosed structural sanity check Chapter 13 §13.3 established.

Real, measured extraction result. A live query of 3,000 raw CHEMBL203 IC50 records curated down to 1,508 real, distinct, standardized compounds (860 measured active / 648 inactive at the 1 µM convention Chapter 7 established) — Section 16.2.2 trains Stage 2's QSAR model on exactly this real dataset.

16.2.2 Stage 2 — Real QSAR model and ADMET filter

A message-passing neural network — Chapter 6's own `NNConv`-based MPNN architecture (Gilmer et al., 2017), reused with zero architectural changes — is retrained from scratch as a real pIC50 regressor on this chapter's own curated EGFR dataset, evaluated under a real Bemis-Murcko scaffold split (Bemis & Murcko, 1996) rather than a random split, for the same honest, not-inflated-by-analog-leakage reason Chapters 5-7 each already established for this exact kind of structure-activity dataset. Every generated candidate is additionally screened by Chapter 13's own real Tier 1 rule-based filter unchanged — Lipinski Ro5, Veber's rules, RDKit's PAINS catalog, and a QED floor — providing the drug-likeness half of Stage 3's reward signal alongside the QSAR model's own predicted potency.

```
class MPNNRegressor(nn.Module):
    def __init__(self, hidden_dim=64, num_layers=3):
        super().__init__()
        self.atom_encoder = AtomEncoder(hidden_dim)
        self.bond_encoder = BondEncoder(hidden_dim)
        self.convs = nn.ModuleList(
            NNConv(hidden_dim, hidden_dim, edge_net(hidden_dim), aggr="mean")
            for _ in range(num_layers)
        )
        self.readout = nn.Sequential(nn.Linear(hidden_dim, hidden_dim),
nn.ReLU(), nn.Linear(hidden_dim, 1))
```

Real, measured QSAR performance. Stage 1's real, 1,508-compound curated dataset is an order of magnitude larger than Chapter 5's own hERG QSAR dataset and Chapter 6's own ESOL/FreeSolv benchmarks — the first time this book has trained a property predictor at genuine drug-discovery-campaign scale rather than a small public benchmark's. Under the real scaffold split (1,206 train / 302 test compounds), the retrained MPNN reaches:

Metric	Value
RMSE (pIC50 units)	1.040
MAE (pIC50 units)	0.749
R^2	0.451
Spearman ρ	0.728

An RMSE just above one full pIC50 unit (roughly a 10× error in predicted IC50) and a Spearman ρ of 0.73 are real, honest, literature-consistent numbers for a scaffold-split MPNN on a real, heterogeneous, multi-assay ChEMBL extraction — a genuinely harder, less curated real dataset than Chapter 5's own single-assay hERG benchmark or Chapter 6's own single-source MoleculeNet benchmarks, and the scaffold split (rather than a random split) means this number is not inflated by near-duplicate analogs straddling the train/test boundary, the same honest-evaluation discipline Chapters 5-7 already established for exactly this failure mode.

16.2.3 Stage 3 — Real generative Transformer + RL, with an upgraded reward oracle

Chapter 7's own real SELFIES-token decoder-only Transformer, pretrained by next-token prediction on real active EGFR compounds and fine-tuned with REINFORCE (Williams, 1992; Olivecrona et al., 2017), is reused architecturally unchanged. What changes is the **reward function**: Chapter 7's reward oracle was a binary XGBoost classifier predicting $P(\text{active})$; this chapter's reward oracle is Stage 2's own continuous MPNN regressor, mapped to a bounded reward via a real, disclosed linear scaling over a sensible medicinal-chemistry potency window (pIC50 $\in [4, 9]$, i.e. 100 μM down to $1\, nM$), combined with the Stage 2 ADMET filter's own pass/fail call:

$$r(\text{SMILES}) = 0.7 \cdot \text{clip}\left(\frac{\widehat{\text{pIC50}} - 4}{9 - 4}, 0, 1\right) + 0.3 \cdot \mathbb{1}[\text{ADMET pass}]$$

This is a genuinely more informative reward signal than Chapter 7's own binary classifier — it distinguishes a barely-active from a highly-potent candidate, information a $P(\text{active})$ classifier collapses to the same reward — while reusing the exact same REINFORCE mechanics (a running-mean baseline, policy-gradient ascent on $\mathbb{E}[r \cdot \log p(\text{sequence})]$) Chapter 7 already validated.

A real bug, found and fixed before any number below was finalized. An MPNN regressor is trained to predict a *normalized* target, $(\text{pIC50} - \mu)/\sigma$, and only converted back to real pIC50 units inside `train_qsar_model`'s own internal evaluation loop. This chapter's first working version of the standalone scoring function `predict_pic50` — the function Stage 3's reward and Stage 4's candidate ranking both call directly — omitted that same conversion, silently returning the raw normalized model output as if it were already a real pIC50. The bug did not crash anything: it produced plausible-looking Python floats, just on the wrong scale (systematically around $-\mu/\sigma \approx -4.3$ pIC50 units too low), so the pipeline ran to completion and produced results that looked superficially real. It was caught by exactly the kind of check this book has run before trusting any new predictive step (Chapter 13 §13.3's replicate-deduplication bug was caught the same way): predicting on a handful of real *training* molecules and comparing against their own real, known pIC50 values, rather than trusting the pipeline's internal consistency alone. The real, observed effect on Stage 3, before the fix: RL made essentially zero real progress (mean predicted pIC50 moved from -2.961 to -2.934 over 25 iterations) because every generated molecule's activity-reward term was floor-clipped at 0 regardless of relative quality, leaving only the ADMET term for REINFORCE to actually optimize. Fixed by attaching the real training mean/std to the trained model object and applying them inside `predict_pic50` (`ch16_capstone_oncology/tests/test_oncology_capstone.py`'s `test_predict_pic50_denormalizes_using_the_models_own_training_statistics` is the regression test this added); every number from here on reflects the corrected pipeline.

Real, measured generative results. Pretraining on the real 842 active EGFR compounds (vocabulary size 42) for 20 epochs, then 25 real REINFORCE iterations against the corrected reward:

Quantity	Value
Mean predicted pIC50, pretrained-only policy	1.932
Mean predicted pIC50, after RL fine-tuning	2.122
Final mean reward	0.295
Valid fraction (post-RL)	0.995
Unique fraction of valid (post-RL)	0.990
Novel fraction of valid (post-RL)	1.000

RL moved the real mean predicted pIC50 by $+0.19$ units — a real, modest, honestly-reported improvement (roughly a 1.5× shift in predicted potency), not a dramatic one, over only 25 real REINFORCE iterations (Chapter 7's own default budget, reused unchanged). Every one of the sampled valid molecules is novel by exact-match against the real training set, and nearly all are mutually unique — the policy is not collapsing onto a small set of memorized or repeated outputs.

16.2.4 Stage 4 — Real docking and MD verification of the pipeline's own generated candidates

Every real candidate this capstone docks is novel — generated by Stage 3, never present in the real ChEMBL training data — a real, qualitative difference from Chapter 11's own hands-on project, which docked a real, pre-existing benchmark library. The receptor preparation, focused-box protocol, and Vina settings are Chapter 11's own, reused unchanged (pocket-informed 22.5 Å cube centered on erlotinib's real crystallographic centroid, exhaustiveness 8, and the same Windows-CLI-binary fallback Chapter 11's own feasibility investigation established for this pip-first, conda-free authoring environment — DiffDock's own infeasibility finding, established once in Chapter 11 §11.2 and reused directly by Chapters 13 and 16 alike, is not re-investigated here). A real redocking validation control — erlotinib docked back into its own real 1M17 pocket — is run first, the same self-consistency check Chapters 11 and 13 each ran on their own receptor before trusting it:

Replicate	Affinity (kcal/mol)	RMSD to crystal pose (Å)	Correct pose (<2.0 Å)?
1	−7.736	2.623	No
2	−7.725	2.610	No
3	−7.451	1.635	Yes
Mean	**-7.637**	**2.289**	**1/3**

This real result reproduces Chapter 11′s own finding on this exact receptor/ligand pair almost exactly (Chapter 11 §11.4 reported erlotinib's own real redocking RMSD "consistently landing just above" the 2.0 Å threshold, 0/5 replicates strictly under it) — a real, reassuring cross-validation that this chapter's independently-run receptor preparation reproduces Chapter 11′s own established result on the same real structure, not evidence of a docking-protocol problem specific to this chapter.

Real docking results. The top 8 real, novel, ADMET-passing candidates by predicted pIC50 were docked; 7 succeeded (one, `GEN_001`, failed 3D conformer embedding — a real, disclosed per-compound failure, not silently dropped):

Molecule	Predicted pIC50	Vina affinity (kcal/mol)	Contains Br?
GEN_006	5.108	−8.452	Yes
GEN_002	6.149	−7.253	Yes
GEN_004	5.524	−6.680	Yes
GEN_000	6.631	−6.519	Yes
GEN_003	5.986	−5.413	No
GEN_007	5.062	−5.180	No
GEN_005	5.360	−2.674	No

A real, disclosed halogen-coverage gap, caught and corrected before finalizing. The four most favorably-docked real candidates all contain bromine — a real, chemically unsurprising outcome (halogens are common, real, potency-enhancing substituents; several real approved EGFR inhibitors carry them too) — but bromine falls outside ANI-2x's own real trained element coverage (H, C, N, O, F, Cl, S; Chapter 12 §12.3). A first version of this stage simply ran MD on the top 3 candidates by docking affinity, discovering the coverage failure per compound only after the fact — and because all top 3 happened to be brominated, that version produced *zero* real MD trajectories, an uninformative result this chapter does not report. The corrected, disclosed selection criterion instead checks real element compatibility *before* ranking: the top 3 real, ANI-2x-compatible candidates by docking affinity are the ones actually simulated (`ani2x_compatible_elements`, tested directly in this chapter's own test suite). All three real, short ligand-alone trajectories are stable:

Molecule	Vina affinity (kcal/mol)	Atoms	Mean RMSD (Å)	Max RMSD (Å)	Stable?
GEN_003	−5.413	25	0.551	0.894	Yes
GEN_007	−5.180	35	0.938	1.365	Yes
GEN_005	−2.674	9	0.774	1.361	Yes

All three real trajectories are stable by this book's established bounded-RMSD criterion (Chapters 12-13), and — like every prior chapter's own short ligand-alone MD — this is a real integration-correctness/gross-stability check, not a converged binding-affinity claim. The four brominated, better-docked real candidates were not left unexamined by mistake; they are honestly reported as docked-but-not-MD-verified in this run, a real, disclosed scope limit of ANI-2x's own real element coverage rather than a result quietly worked around.

Real, measured wall-clock cost. The real docking of 7 candidates took 432.2 s total (mean 61.7 s/compound); the real MD of 3 candidates took 184.5 s total — 616.7 s (about 10.3 minutes) of real Stage 4 compute altogether, well under Chapter 11′s own measured ~9-10 minutes-*per-compound* rate on its larger, chemically diverse ChEMBL benchmark. The real, disclosed reason is not a different protocol but a different candidate population: this stage's own real generated molecules (Section 16.2.3) are, on average, smaller and structurally simpler than Chapter 11′s real, diverse benchmark set, and Vina's own search cost scales with a ligand's real rotatable-bond count and size — a real, direct illustration of the same speed-accuracy trade-off Chapter 11 §11.3 introduced as theory, observed here as a real, measured side effect of a generative pipeline's own candidate distribution rather than deliberately tuned for speed.

16.2.5 Stage 5 — Automated technical report

`generate_report` assembles every real number the pipeline produced — target information, QSAR performance, before/after-RL generative statistics, the redocking control, and a per-candidate table of predicted activity, real docking affinity, and real MD stability — into a single, self-contained, real HTML file (`results/technical_report.html`), with each candidate's real 2D structure rendered directly by RDKit and embedded as a base64 PNG (no external image hosting, no JavaScript, opens in any browser offline). This is a genuinely new kind of deliverable for this book: every prior chapter's own results appear in `chapter.md`'s prose, authored by hand from a real `results/*.json` file; this stage's report is itself *generated code*, run as part of the pipeline rather than written after the fact — the literal, concrete form the outline's "automated technical report" language calls for.

Limitations and what comes next

This capstone chains five real stages into one working pipeline and reports every real number honestly, including the ones that did not go as originally planned: the reward-scaling bug Section 16.2.3 discloses in full, and the halogen/ANI-2x coverage gap Section 16.2.4 discloses in full, are both real defects this project's own execution surfaced and fixed before any number here was finalized — the same "catch it, fix it, add a regression test, disclose it" discipline Chapter 13′s own replicate-deduplication bug established. Real, disclosed scope limits remain even after those fixes. The QSAR model's own R^2 of 0.45 means Stage 3′s reward signal, however more informative than Chapter 7′s binary classifier, is still a real, imperfect proxy for true EGFR affinity — exactly the kind of oracle imperfection Chapter 7 §7.5 already discussed as a fundamental limit of reward-shaped generative design, not a defect specific to this chapter's own model. The four best-docked real candidates were never verified by real MD in this run, purely a real tooling-coverage gap rather than a judgment that they are unstable. And every docking and MD number in Section 16.2.4 describes a real, computed pose and a real, short trajectory — not a wet-lab measurement; the real, final verification this pipeline's own top candidates would need before any synthesis decision is exactly the kind of real experimental follow-up no computational pipeline, this one included, can substitute for. Chapter 17 shifts this book's focus to its second and final capstone — a real, complete de novo antibody design pipeline against a viral surface antigen, chaining Part III's own protein-design methods (RFdiffusion, ProteinMPNN, AlphaFold3/ESMFold) the same way this chapter chained Parts I-IV's small-molecule methods.

References

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See Chapter 1′s references for Lipinski et al. (2001, Rule of Five) and Mendez et al. (2019, ChEMBL); Chapter 2′s for Krenn et al. (2020, SELFIES); Chapter 5′s for Bemis & Murcko (1996, scaffold definition); Chapter 7′s for Vaswani et al. (2017, Transformer); Chapter 11′s for Stamos et al. (2002, also listed above), O'Boyle et al. (2011, OpenBabel), and the DiffDock feasibility investigation reused directly in Section 16.2.4; Chapter 12′s for Eastman et al. (2017, OpenMM); and Chapter 13′s for Veber et al. (2002), Baell & Holloway (2010, PAINS), and Bickerton et al. (2012, QED) — all reused here rather than re-listed.

All dataset sizes, QSAR metrics, generative-model statistics, docking results, and MD stability numbers cited in Section 16.2 were computed directly by running `oncology_capstone.py` end to end on 2026-08-22, not taken from a secondary source — see `results/capstone_results.json` and `results/technical_report.html` to reproduce.